Reg. No									

St. Joseph's College of Arts & Science (Autonomous)
St. Joseph's College Road, Cuddalore – 607001

CS102 – DIGITAL LOGIC FUNDAMENTALS

Time : 3 hrs **Max Marks :75**

SECTION – A (5X5=25)

Answer any **FIVE** Questions

1. Write common postulates used in Boolean algebra.
2. Write note on (i) X-OR gate (ii) NOR gate
3. Write note on product of sums.
4. Explain the uses of Don't care conditions with K-map.
5. Define Adder. Explain the working methodologies of Full-adder.
6. Explain the functions of seven segment decoders.
7. Write short note on counter.
8. What is shift register? Explain.

SECTION – B (5X10=50)

Answer any **FIVE** Questions

9. What are the different binary codes used in binary number system? Explain.
10. Simplify the Boolean function:
F(A,B,C,D) = $\sum$ (0,2,4,6,9,11,13,15,17,21,25,27,29,31)

Using K-Map.

11. Why do you use ROM/PLA? Explain.

12. Write note on (i) Half-Subtractor (ii) Full-Subtractor

13. Draw the circuit diagram of Demuliplexers. Explain their functions.

14. Explain the uses of Encoder.

15. Write note on (i) D Flip-flop (ii) T Flip-flop

16. Explain ALU.

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